Quantum Darwinism is not enough: Consistency of facts in Wigner's friend scenarios

(Dated: January 4, 2026)

I. MODEL

We consider a composite quantum system composed of three distinct parts:

1. a system qubit S,
2. a quantum observer (the Friend) F, represented by a memory register,
3. an environment E consisting of N qubits $E = \{E_1, E_2, \ldots, E_N\}$.

The total Hilbert space is given by

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_F \otimes \bigotimes_{i=1}^{N} \mathcal{H}_{E_i}. \tag{1}$$

We assume an initially uncorrelated pure state,

$$|\Psi_0\rangle = |+\rangle_S \otimes |0\rangle_F \otimes |0\rangle_E^{\otimes N}, \tag{2}$$

where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \tag{3}$$

is a coherent superposition of the system pointer basis states.

This choice represents a minimal setting in which coherence, measurement, and environmental monitoring can all be treated within a fully unitary framework.

II. FRIEND MEASUREMENT AS UNITARY PREMEASUREMENT

The measurement performed by the Friend is modeled as a unitary premeasurement interaction between the system and the Friend's memory. Specifically, we consider the controlled-NOT–type unitary

$$U_{SF} = |0\rangle\langle 0|_S \otimes \mathbb{I}_F + |1\rangle\langle 1|_S \otimes X_F, \tag{4}$$

where X is the Pauli-X operator acting on the Friend's memory qubit.

Applying this interaction to the initial state yields

$$|\Psi_{SF}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_S |0\rangle_F + |1\rangle_S |1\rangle_F \right). \tag{5}$$

This state represents perfect correlation between the system and the Friend's memory while remaining fully coherent. Importantly, no projection or collapse postulate is invoked. From the Friend's internal perspective, the memory register stores a definite outcome, whereas from an external perspective the joint system remains in an entangled superposition.

III. ENVIRONMENT MONITORING

Following the Friend's measurement, the environment interacts with the system through a sequence of controlled-unitary operations,

$$U_{SE} = \prod_{i=1}^{N} \left(|0\rangle\langle 0|_S \otimes \mathbb{I}_{E_i} + |1\rangle\langle 1|_S \otimes U_{E_i} \right), \tag{6}$$

where U_{E_i} is a single-qubit unitary acting on the ith environment qubit.

These interactions imprint information about the system's pointer states into the environment. After all interactions, the global state takes the form

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_S |0\rangle_F |E_0\rangle + |1\rangle_S |1\rangle_F |E_1\rangle \right), \tag{7}$$

where $|E_0\rangle$ and $|E_1\rangle$ denote environment states that become increasingly distinguishable as the number of environment qubits grows.

This structure underlies the mechanism of Quantum Darwinism.

IV. QUANTUM DARWINISM

To quantify the information shared between the system and a fragment of the environment $E_f \subset E$, we use the quantum mutual information

$$I(S : E_f) = H(S) + H(E_f) - H(SE_f), \tag{8}$$

where $H(\rho) = -\operatorname{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy.

Quantum Darwinism predicts the emergence of a redundancy plateau, characterized by

$$I(S : E_f) \approx H(S) \tag{9}$$

for many disjoint fragments E_f whose size is much smaller than the full environment.

The redundancy R_δ is defined as the maximum number of disjoint fragments satisfying

$$I(S : E_f) \geq (1 - \delta) H(S), \tag{10}$$

where δ quantifies the tolerated information deficit.

V. WIGNER'S COHERENCE TEST

From Wigner's perspective, the combined system SF may remain coherent. To test this coherence operationally, Wigner performs a joint measurement on S and F using the projector

$$W = \left|\Phi^+\right\rangle\!\left\langle\Phi^+\right|_{SF}, \qquad \left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \tag{11}$$

The expectation value

$$\langle W \rangle = \operatorname{Tr}(W \rho_{SF}) \tag{12}$$

distinguishes a coherent entangled state from a classical mixture and serves as a witness of observer-accessible quantum coherence.

VI. CONSISTENCY GAP

For an environment fragment E_f, we define the *consistency gap*

$$\Delta_{\text{cons}}(E_f) = I(F : E_f) - \chi(F : E_f), \tag{13}$$

where $\chi(F : E_f)$ is the Holevo information,

$$\chi(F : E_f) = H\left(\sum_f p_f \rho_{E_f|f} \right) - \sum_f p_f H(\rho_{E_f|f}), \tag{14}$$

representing the maximum classically accessible information about the Friend's memory contained in the environment fragment.

A nonzero consistency gap indicates the presence of correlations that are not operationally accessible as classical records, signaling a breakdown of observer-independent facts.